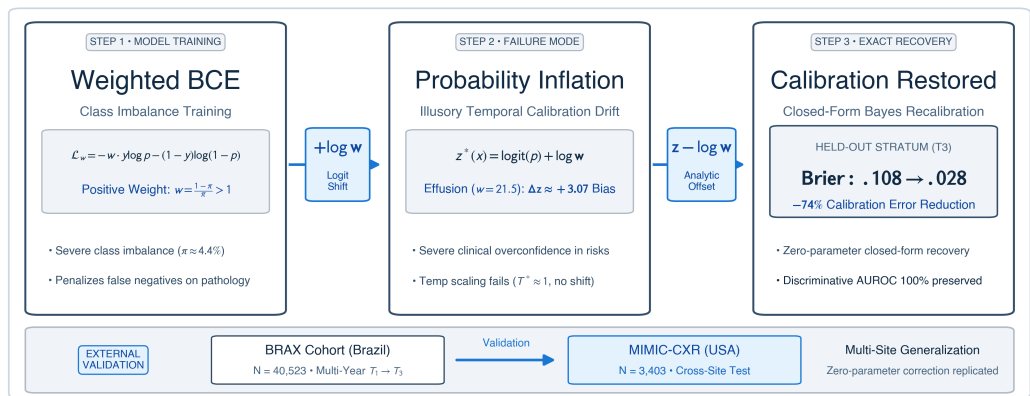


Graphical Abstract

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

Vishesh Panghal



Highlights

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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- Positive loss weighting induces an additive population logit offset of $\log w$.
- Finite neural network logits empirically cluster tightly around theoretical $\log w$.
- Temperature scaling cannot correct this intercept-type calibration bias.
- Analytic offset ($z - \log w$) cuts 74% of Brier error without retraining.
- Selective triage exposes trade-offs between calibrated risk and screening referral.

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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Abstract

Background and Objective: Clinical deep learning models for chest radiography frequently experience severe probability miscalibration under distribution shift. While positive-weighted binary cross-entropy (BCE) is standard practice to mitigate acute class imbalance, its mathematical distortion of output logit distributions has been conflated with feature drift. This study investigates the algebraic mechanism of loss-induced logit distortion, evaluates why standard post-hoc calibration fails, and formulates a zero-retraining recalibration and selective deferral pipeline.

Methods: We conducted a controlled 2×2 factorial experiment using two deep convolutional neural network architectures (DenseNet-121, ResNet-50) and two loss functions (unweighted vs. positive-weighted BCE) across three independent random seeds (12 models total) on 40,523 chest radiographs from BRAX. Models were trained on an anchor deployment stratum and evaluated across subsequent deidentified-date strata (T_1, T_2, T_3) with strict patient isolation. Recalibration strategies—temperature scaling, Platt scaling, and an $O(1)$ closed-form analytic logit offset ($z - \ln w$)—were evaluated alongside patient-clustered bootstrap ($B = 2000$). External validity was evaluated on MIMIC-CXR-JPG (3,403 radiographs, 289 patients).

Results: Positive-weighted BCE mathematically induces an additive population logit shift of $\ln w$, which temperature scaling structurally cannot resolve because scaling modifies logit dispersion without adjusting the intercept. In

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empirical finite networks, logit differences clustered tightly around theoretical values (mean $\Delta z = 3.29 \pm 0.81$ vs. theoretical 3.07 for Pleural Effusion). On stratum T_3 , the zero-parameter analytic offset reduced Brier score by 73.7% ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$], $0.1079 \rightarrow 0.0284$) and reduced intercept error from $\alpha = -2.96$ to -0.45 without validation tuning. On MIMIC-CXR-JPG, discrimination was pathology-dependent (AUROC 0.688 for Effusion), with analytic offset reducing Brier error for Cardiomegaly and Pneumonia. Uncertainty-based selective deferral reduced automated error rates to < 0.01 at 70% coverage.

Conclusions: Severe probability miscalibration in chest radiograph classification across held-out deidentified-date strata is largely an artifact of objective function weighting rather than intrinsic dataset drift. The closed-form analytic offset provides an immediate, constant-time $O(1)$ algorithmic recalibration that restores probability reliability across held-out deidentified-date cohorts without model refitting.

Keywords: Chest radiography, Probability calibration, Class imbalance, Distribution shift, Selective prediction, Deep learning, Clinical decision support

1. Introduction

Deep learning models for chest radiograph interpretation have achieved high discriminatory performance across common thoracic abnormalities [1, 2]. However, transferring models from stationary retrospective validation to longitudinal clinical deployment introduces substantial distribution shifts as patient populations, imaging protocols, and disease prevalence evolve over time [3, 4, 5].

While discriminatory ranking (AUROC) often exhibits gradual longitudinal attenuation, high-stakes medical decision support relies fundamentally on *calibrated posterior probabilities*: $P(Y = 1 \mid X = x)$ [6, 7]. Miscalibrated probabilities distort clinical decision cutoffs, degrade expected utility formulations, and compromise automated triage. Recent studies have reported rapid “temporal calibration decay,” asserting that frozen neural networks suffer progressive miscalibration over deployment periods [8, 9]. Concurrently, because thoracic datasets exhibit extreme class imbalance (prevalence often $< 5\%$), models are routinely trained using positive-weighted binary cross-entropy (BCE) [1, 10].

In this work, we demonstrate that these two phenomena are deeply entangled: the prevailing attribution of temporal calibration collapse to intrinsic feature drift is largely an artifact of uncorrected post-training calibration offsets under label distribution shifts [11, 12]. Under population risk minimization, training with positive loss weight $w > 1$ mathematically shifts the Bayes-optimal logit by an additive constant $+\ln w$ [13, 14]. Standard post-hoc calibration methods—most notably temperature scaling [15]—rescale logit variance (z/T) but cannot adjust the logit intercept. Consequently, weighted models continuously over-predict positive probabilities, producing severe Brier score and Expected Calibration Error (ECE) degradation that compounds as disease prevalence fluctuates over time.

1.1. Related Work

Probability Calibration and Class Imbalance: Deep neural networks frequently exhibit over-confidence due to unconstrained cross-entropy minimization [15]. While post-hoc recalibration techniques such as temperature scaling, histogram binning, and Platt scaling are standard [15, 16, 6], they were primarily formulated for balanced distributions. In clinical prediction modeling, Piccininni et al. [17] demonstrated that correcting class imbalance via resampling inherently distorts probability calibration, while plug-in corrections can recover valid absolute risks. In loss-based weighting, King and Zeng [13] established in classical logistic regression that choice-based sampling proportion adjustments induce an intercept shift of $\ln w$. Recently, Caplin, Martin, and Marx [14] analyzed loss-weighted deep learning through an economic lens, proving that class weighting is mathematically incompatible with raw posterior calibration and formulating an inverse machine learning framework to recover undistorted likelihoods on a static pneumonia task. However, their work was confined to single-split evaluations, leaving unexamined how loss-induced logit shifts behave under sequential temporal cohort shifts, whether standard temperature scaling fails, and how post-hoc recalibration performs under cross-institutional transfer.

Distribution Shift and Selective Prediction: Temporal dataset shift encompasses changes in feature distributions $P(X)$, conditional distributions $P(Y | X)$, and label priors $P(Y)$ [3, 12]. Alexandari et al. [11] established that bias-corrected calibration provides robust adaptation under label shift. Selective classification allows models to abstain on ambiguous inputs and route them to expert radiologists [18, 19, 20]. While selective triage studies focus primarily on abstention policies, the interaction between loss-induced

55 calibration distortion and downstream referral triage remains unexplored. In
 56 this work, we employ selective prediction as an operational lens to examine
 57 the real-world clinical trade-offs between calibrated probability estimation
 58 and screening triage.

59 1.2. Contributions

60 To resolve algorithmic loss bias from distribution shift across cohort strata,
 61 this paper provides:

- 62 1. **Controlled 2×2 Experimental Design:** We evaluate two clinical
 63 architectures (DenseNet-121, ResNet-50) and two loss objectives (stan-
 64 dard vs. positive-weighted BCE) across three independent seeds (12
 65 models total) on 40,523 radiographs from BRAX [2], tracking discrim-
 66 ination and calibration across three held-out deidentified-date strata
 67 (T_1, T_2, T_3) with strict patient isolation.
- 68 2. **Mathematical Derivation and Empirical Clustering of Logit**
 69 **Shift:** We prove that positive-weighted BCE introduces a constant
 70 population logit offset $\ln w$ (Section S1 of the Supplementary Material).
 71 Across 16,604 evaluations, finite neural network logits cluster tightly
 72 around theoretical $\ln w$ (mean $\Delta z = 3.29 \pm 0.81$ vs. theoretical 3.07
 73 for Pleural Effusion). Crucially, we prove that temperature scaling
 74 structurally fails to correct this intercept displacement ($\alpha \approx -2.96$),
 75 whereas a closed-form analytic offset ($z - \ln w$) eliminates 73.7% of
 76 stratum T_3 Brier error ($0.1079 \rightarrow 0.0284$) and restores calibration parity
 77 without model retraining.
- 78 3. **Leak-Free Cross-Fitting and Patient-Clustered Bootstrap:** We
 79 implement a 5-fold patient-clustered cross-fitting protocol on T_1 and
 80 evaluate out-of-sample generalization on T_2 and T_3 backed by 2,000-
 81 replicate patient-clustered bootstrap intervals.
- 82 4. **Operational Selective Prediction and Cross-Institutional Trans-**
 83 **portability:** We formalize budget-constrained uncertainty-ranked selec-
 84 tive deferral, quantifying the trade-off between calibrated risk estimation
 85 and screening triage. We stress-test external transportability on MIMIC-
 86 CXR-JPG (3,403 images, 289 patients), delineating the capabilities and
 87 boundaries of zero-retraining recalibration under real-world domain
 88 shift.

Table 1: Cohort Demographics, Radiographic Projections, and Target Condition Prevalence across Stratified Cohorts

Cohort Stratum	Images (N)	Patients (N)	Studies (N)	Sex (% M / F)	Age (Mean $\pm$ SD)	Views (% PA / AP / Lat / Unk)	Pleural Effusion N (%)	Cardiomegaly N (%)	Pneumonia N (%)	Edema N (%)
Train (Pre-deployment)	25,123	11,265	14,315	49.3% / 50.7%	46.0 $\pm$ 24.3	34.0% / 13.5% / 0.0% / 27.2%	1,116 (4.44%)	2,398 (9.55%)	456 (1.82%)	23 (0.09%)
Val (Pre-deployment)	7,098	3,281	4,012	49.8% / 50.2%	46.2 $\pm$ 24.1	34.7% / 12.2% / 0.0% / 27.7%	295 (4.16%)	681 (9.59%)	128 (1.80%)	8 (0.11%)
Test T1 (2015)	1,879	913	1,039	47.8% / 52.2%	44.5 $\pm$ 23.9	34.4% / 10.2% / 0.0% / 29.0%	147 (7.82%)	147 (7.82%)	50 (2.66%)	4 (0.21%)
Test T2 (2016)	3,987	1,832	2,249	48.5% / 51.5%	46.1 $\pm$ 23.8	34.6% / 12.0% / 0.0% / 27.6%	177 (4.44%)	422 (10.58%)	79 (1.98%)	13 (0.33%)
Test T3 (2017)	2,436	1,164	1,382	50.9% / 49.1%	46.0 $\pm$ 24.6	34.4% / 12.4% / 0.0% / 28.1%	85 (3.49%)	258 (10.59%)	50 (2.05%)	0 (0.00%)
Test All (Pooled)	8,302	3,807	4,670	49.0% / 51.0%	45.7 $\pm$ 24.1	34.5% / 11.7% / 0.0% / 28.1%	361 (4.35%)	827 (9.96%)	179 (2.16%)	17 (0.20%)

2. Materials and Methods

2.1. Dataset and Cohort Stratification

We utilize the Brazilian Chest X-ray (BRAX) v1.1.0 benchmark [2]. BRAX contains 40,967 radiographs labeled with 14 thoracic findings extracted from Portuguese reports via a validated NLP pipeline. After excluding 444 radiographs from 89 patients whose examinations crossed partition boundaries, the analytic cohort comprised 40,523 radiographs acquired at Hospital Israelita Albert Einstein in São Paulo, Brazil.

In compliance with HIPAA de-identification standards, timestamps are protected as fictitious 8-digit calendar integers (YYYYMMDD) preserving intra-patient acquisition intervals. Because cross-patient calendar synchronization is unverified, partitions represent sequential deidentified-date strata rather than synchronized calendar years. Studies are partitioned chronologically around a strict model freeze boundary ($T_{\text{freeze}} = 2015-07-01$): training studies before 2013-09-01 ($N = 25,123$ images, 11,265 patients, 14,315 studies); validation studies between 2013-09-01 and 2015-06-30 ($N = 7,098$ images, 3,281 patients, 4,012 studies); and held-out test studies between 2015-07-01 and 2017-12-17 ($N = 8,302$ images, 3,807 patients, 4,670 studies) partitioned into three consecutive strata: T_1 ($N = 1,879$), T_2 ($N = 3,987$), and T_3 ($N = 2,436$). Strict patient isolation is enforced ($\text{Patients}_{\text{train}} \cap \text{Patients}_{\text{val}} \cap \text{Patients}_{\text{test}} = \emptyset$). Complete cohort characteristics are summarized in Table 1.

2.2. External Stress-Testing Cohort (MIMIC-CXR-JPG)

To assess whether the mathematical calibration mechanism survives cross-institutional domain shift, frozen BRAX models were externally evaluated on all eligible frontal radiographs (AP and PA views) from the official patient-partitioned MIMIC-CXR-JPG test split [21, 22]. The cohort comprises $N = 3,403$ frontal radiographs ($N = 3,041$ diagnostic studies across $N = 289$ unique patients; AP: 2,405 [70.67%], PA: 998 [29.33%]). Ground-truth labels were extracted using the official CheXpert labeler under the uncertainty-zero (U -zero) policy: Pleural Effusion 32.18% ($N = 1,095$ positive radiographs

across 178 positive and 273 negative patients; 990 studies), Cardiomegaly 26.33% ($N = 896$ across 183 positive and 282 negative patients; 808 studies), Edema 21.33% ($N = 726$ across 150 positive and 287 negative patients; 659 studies), and Pneumonia 10.05% ($N = 342$ across 137 positive and 286 negative patients; 309 studies). Patient-clustered non-parametric bootstrap inference ($B = 1,000$ replicates) accounts for repeated within-patient examinations and longitudinal disease transitions across serial studies. No model fine-tuning or retraining was performed.

2.3. Architectures and Training Objectives

We evaluate two standard clinical vision architectures initialized from ImageNet pre-trained weights:

- **DenseNet-121** [23]: Employs dense feature concatenation, standard for thoracic pathology benchmarks [1].
- **ResNet-50** [24]: Employs residual skip connections as an independent structural comparator.

Each architecture is trained under two distinct loss objectives:

1. Unweighted Binary Cross-Entropy (Standard BCE):

$$\mathcal{L}_{\text{std}}(y, \hat{p}) = -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (1)$$

where $\hat{p} = \sigma(z) = 1/(1 + e^{-z})$ is the sigmoid probability from raw logit z .

2. Positive-Weighted Binary Cross-Entropy (Weighted BCE):

$$\mathcal{L}_{\text{pos}}(y, \hat{p}) = -[w \cdot y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (2)$$

where $w = (N_{\text{total}} - N_{\text{pos}})/N_{\text{pos}}$ represents the inverse positive prevalence in the training set (e.g., $w = 21.51$ for Pleural Effusion, $w = 9.48$ for Cardiomegaly, $w = 54.09$ for Pneumonia).

Models are trained with the AdamW optimizer (learning rate 10^{-4} , weight decay 10^{-2}) and a Cosine Annealing schedule over 15 epochs (batch size 64, AMP fp16, 512×512 resolution). Pleural Effusion was pre-specified as the primary target pathology due to its stable evaluability across all longitudinal held-out strata ($T_1 = 1,879, T_2 = 3,987, T_3 = 2,436$); Cardiomegaly, Pneumonia, and Edema served as secondary mechanistic validation targets. Checkpoints were selected by highest validation AUROC.

Three random seeds ($S \in \{42, 123, 2026\}$) were trained per configuration (12 models total). For unified benchmarking, a 3-seed probability ensemble was constructed by averaging predicted probabilities: $\bar{p}_{\text{raw}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x))$ and $\bar{p}_{\text{corrected}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x) - \ln w)$. The empirical logit offset was evaluated across all 16,604 ensemble-paired evaluations (8,302 images $\times$ 2 architectures; 49,812 seed-level pairs):

$$\Delta z(x) = \bar{z}_w(x) - \bar{z}_u(x) = \frac{1}{3} \sum_{s \in S} (z_{w,s}(x) - z_{u,s}(x)) \quad (3)$$

2.4. Mathematical Mechanics of Loss Weighting and Calibration

Under positive-weighted binary cross-entropy, the conditional expected risk with positive weight $w > 0$ is:

$$\mathbb{E}_{Y|X}[\mathcal{L}_{\text{pos}}(Y, \hat{p})] = -w \cdot P(Y = 1 | X) \ln \hat{p} - P(Y = 0 | X) \ln(1 - \hat{p}) \quad (4)$$

Setting the first-order condition $\frac{\partial}{\partial \hat{p}} \mathbb{E}[\mathcal{L}_{\text{pos}}] = 0$ yields the optimal predicted probability:

$$\hat{p}^*(x) = \frac{w \cdot P(Y = 1 | X = x)}{w \cdot P(Y = 1 | X = x) + P(Y = 0 | X = x)} \quad (5)$$

Converting to log-odds yields the Bayes-optimal logit:

$$z^*(x) = \ln \left(\frac{P(Y = 1 | X = x)}{P(Y = 0 | X = x)} \right) + \ln w = z_{\text{true}}(x) + \ln w \quad (6)$$

(see Section S1 of the Supplementary Material for the complete algebraic proof). Equation (6) derives that positive loss weighting shifts the model’s population Bayes-optimal logit by $+\ln w$, connecting classical choice-based sampling corrections [13] with the class-weight likelihood recovery formulation of Caplin et al. [14]. In finite neural networks, empirical logits fluctuate around this theoretical attractor.

2.4.1. Limitation of Temperature Scaling

Temperature scaling [15] adjusts probabilities via a single scalar T : $p_{\text{temp}} = \sigma(z/T)$. In the Cox calibration framework [6, 7]:

$$\text{logit}(P(Y = 1 | X)) = \alpha + \beta \cdot \text{logit}(\hat{p}) \quad (7)$$

where $\alpha = 0$ represents perfect calibration-in-the-large and $\beta = 1$ denotes ideal slope. Because positive weighting induces an additive shift $z \approx z_{\text{true}} + \ln w$, the uncorrected logit suffers an acute intercept discrepancy ($\alpha \approx -\ln w \ll 0$). Temperature scaling multiplies logits by $1/T$, adjusting slope $\beta \rightarrow \beta \cdot T$, but is mathematically incapable of adjusting the additive constant α . In low-prevalence clinical domains where negative cases dominate, the optimizer is constrained by conflicting slope and intercept pressures, typically selecting $T^* \approx 1.0$ without resolving the underlying intercept displacement.

2.4.2. Post-Hoc Recalibration Suite and Computational Complexity

We evaluate four post-hoc recalibration functions:

1. **Temperature Scaling (T^*):** $p = \sigma(z/T^*)$, optimized via bounded Brent minimization on $[0.01, 50.0]$ with an explicit monotonicity constraint; incurs $O(N_{\text{val}})$ optimization and $O(1)$ inference.
2. **Analytic Offset ($z - \ln w$):** Directly subtracts the theoretical logit shift: $p = \sigma(z - \ln w)$. Requires zero validation data and zero numerical optimization ($O(1)$ training and inference), resetting $\alpha \approx 0$ by construction while preserving model discrimination and slope.
3. **Fitted Offset ($z + a^*$):** Optimizes a scalar intercept $a^* \in [-10.0, 10.0]$ on validation log-likelihood, holding slope fixed at 1.0 ($O(N_{\text{val}})$ fitting, $O(1)$ inference).
4. **Platt Scaling ($a + bz$):** Logistic regression on logits with a non-negative slope constraint ($b \geq 0$) ($O(k \cdot N_{\text{val}})$ fitting, $O(1)$ inference).

2.5. Cross-Fitting and Resampling Evaluation Protocol

On deployment stratum T_1 (2015-H2), calibrator parameters are estimated using **5-fold patient-clustered cross-fitting** (all studies from any given patient assigned to a single fold). Calibrators fit on T_1 are frozen and applied strictly out-of-sample on later strata T_2 (2016) and T_3 (2017). All 95% confidence intervals are computed using $B = 2,000$ patient-clustered bootstrap resamples with replacement ($B = 1,000$ for external MIMIC validation). Paired calibrator contrasts are evaluated on identical bootstrap resamples.

2.6. Selective Prediction and Deferral Policy

In operational deployment, models can selectively defer ambiguous examinations to expert radiologists [18, 19, 20]. Rather than developing novel

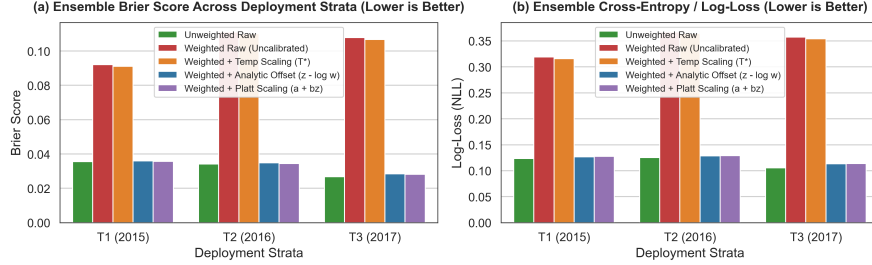


Figure 1: Multi-calibrator evaluation on Pleural Effusion for the DenseNet-121 3-seed ensemble across sequential deidentified-date strata (T_1, T_2, T_3), directly visualizing Table 3. (a) Brier score and (b) Log-loss. Temperature scaling fails to reduce probability error in weighted models, whereas Analytic Offset ($z - \log w$) and Platt Scaling reduce Brier score by 73.7% ($\Delta\text{Brier} = -0.0795$), matching unweighted models.

rejection algorithms, our selective prediction formulation evaluates how loss weighting and calibration impact clinical referral efficiency and automated error rates.

Operating decision cutoffs t^* are determined strictly on development stratum T_1 using Youden’s J index ($t^* \approx 0.014$ for unweighted, $t^* \approx 0.022$ for analytic offset, $t^* \approx 0.33$ for raw weighted models). We define a threshold-relative uncertainty metric based on piecewise distance from t^* :

$$u(x) = 1 - \frac{|\hat{p}(x) - t^*|}{\mathbb{I}(\hat{p}(x) \geq t^*)(1 - t^*) + \mathbb{I}(\hat{p}(x) < t^*)t^*} \in [0, 1] \quad (8)$$

where $u(x) \rightarrow 1$ indicates maximal ambiguity ($\hat{p}(x) \approx t^*$) and $u(x) \rightarrow 0$ indicates decisive confidence. At coverage budget $C \in [0.5, 1.0]$, the system retains the fraction C of cases with lowest uncertainty ($N_{\text{auto}} = \lceil C \cdot N \rceil$) and refers the remaining $1 - C$ to radiologist review ($N_{\text{ref}} = N - N_{\text{auto}}$).

We track triage outcomes across five distinct metrics: (i) retained positives ($N_{\text{pos, ret}}$); (ii) automated true positives ($TP_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 1)$); (iii) automated false negatives ($FN_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 0)$, where $TP_{\text{auto}} + FN_{\text{auto}} = N_{\text{pos, ret}}$); (iv) retained sensitivity ($\text{Sens}_{\text{ret}} = TP_{\text{auto}} / N_{\text{pos, ret}}$); and (v) full-cohort triage sensitivity ($\text{Sens}_{\text{triage}} = 1 - FN_{\text{auto}} / N_{\text{pos, total}}$), assuming deferred cases are correctly diagnosed by expert radiologists.

Algorithm 1 Budget-Constrained Uncertainty-Ranked Selective Deferral Pipeline

Require: Cohort radiographs $\{x_i\}_{i=1}^N$, trained neural network f_θ , positive loss weight w , clinical operating threshold t^* , target coverage budget $C \in (0, 1]$.

Ensure: Automated decisions $\{\hat{y}_i : i \in \mathcal{S}\}$ and referred examination set $\mathcal{D}$.

- 1: **Phase 1: Analytic Recalibration and Uncertainty Scoring**
 - 2: **for** $i = 1$ **to** N **do**
 - 3: Compute raw network logit: $z_i \leftarrow f_\theta(x_i)$
 - 4: **if** model was trained with positive class weight $w > 1$ **then**
 - 5: Apply constant-time closed-form offset: $\tilde{z}_i \leftarrow z_i - \ln(w)$
 - 6: **else**
 - 7: $\tilde{z}_i \leftarrow z_i$
 - 8: **end if**
 - 9: Compute calibrated probability: $\hat{p}_i \leftarrow \sigma(\tilde{z}_i) = \frac{1}{1 + \exp(-\tilde{z}_i)}$
 - 10: Compute threshold-relative normalizer:
 - 11: $s_i \leftarrow \begin{cases} 1 - t^*, & \text{if } \hat{p}_i \geq t^* \\ t^*, & \text{otherwise} \end{cases}$
 - 12: Compute normalized threshold-proximity uncertainty:
 - 13: $u_i \leftarrow 1 - \min\left(1, \frac{|\hat{p}_i - t^*|}{s_i}\right) \in [0, 1]$
 - 14: **end for**
 - 15: **Phase 2: Budget-Constrained Triage Partitioning**
 - 16: Compute automation quota: $k \leftarrow \min(N, \max(1, \lceil C \cdot N \rceil))$
 - 17: Sort cohort indices by ascending uncertainty: $\pi \leftarrow \text{argsort}(\{u_1, \dots, u_N\})$
 - 18: Partition cohort into automated and deferred sets:
 - 19: $\mathcal{S} \leftarrow \{\pi(1), \pi(2), \dots, \pi(k)\}$ {Retained automated cohort}
 - 20: $\mathcal{D} \leftarrow \{\pi(k+1), \dots, \pi(N)\}$ {Referred to expert radiologist review}
 - 21: **Phase 3: Decision Assignment**
 - 22: **for** $i \in \mathcal{S}$ **do**
 - 23: Assign automated binary classification: $\hat{y}_i \leftarrow \mathbb{I}(\hat{p}_i \geq t^*)$
 - 24: **end for**
 - 25: **return** Automated decisions $\{\hat{y}_i\}_{i \in \mathcal{S}}$, deferred cases $\mathcal{D}$.
-

222 3. Experimental Results

223 3.1. Experimental Matrix: Discrimination vs. Calibration Trajectories

224 Table 2 presents the comprehensive evaluation of the 2×2 experimental
225 design comprising two architectures and two loss objectives (with three
226 independently trained random seed replications per configuration, 12 models
227 total) for Pleural Effusion, Cardiomegaly, and Pneumonia across chronological
228 strata T_1, T_2, T_3 .

229 3.1.1. Discrimination Attenuation vs. Calibration Trajectories

230 Both architectures exhibit moderate discriminatory attenuation across
231 subsequent deidentified-date cohort strata. For DenseNet-121 on Pleural
232 Effusion, AUROC decreases from 0.9335 ± 0.0052 at T_1 to 0.8731 ± 0.0120 at T_3
233 ($\Delta\text{AUROC} = -0.0604$); ResNet-50 exhibits an identical downward trajectory
234 ($0.9295 \pm 0.0113 \rightarrow 0.8608 \pm 0.0190$, $\Delta\text{AUROC} = -0.0687$; Table 2). This
235 confirms that discriminatory attenuation occurs gradually across subsequent
236 cohort strata regardless of loss objective.

237 In stark contrast, probability calibration diverges sharply by loss objec-
238 tive. For unweighted models, Brier scores are low and actually improve as
239 prevalence falls ($0.0369 \pm 0.0029 \rightarrow 0.0279 \pm 0.0009$ on Pleural Effusion). For
240 positive-weighted models, Brier scores are severely elevated and appear to
241 deteriorate over time ($0.1081 \pm 0.0394 \rightarrow 0.1254 \pm 0.0456$ on DenseNet-121;
242 $0.1256 \pm 0.0325 \rightarrow 0.1392 \pm 0.0310$ on ResNet-50), with ECE exceeding 22%.
243 Tracking the excess penalty $\Delta\text{Brier} = \text{Brier}_{\text{weighted}} - \text{Brier}_{\text{unweighted}}$ across
244 sequential strata reveals the operational mechanism: when prevalence declines
245 (Pleural Effusion: 5.27% $\rightarrow$ 3.49%), the excess penalty expands ($+0.0712 \rightarrow$
246 $+0.0975$); when prevalence rises (Cardiomegaly: 7.82% $\rightarrow$ 10.59%), it atten-
247 uates ($+0.0729 \rightarrow +0.0683$); and when prevalence is steady (Pneumonia:
248 2.66% $\rightarrow$ 2.05%), it remains fixed ($+0.1079 \rightarrow +0.1112$). A static training
249 logit shift produces surplus false-positive probability mass that is penalized
250 more severely when deployment disease prevalence drops, expanding the
251 apparent temporal calibration gap.

252 3.2. Multi-Calibrator Comparison

253 Table 3 and Fig. 1 benchmark post-hoc calibrators on Pleural Effusion
254 across 2,000 patient-clustered bootstrap resamples on the 3-seed ensemble.

255 For weighted DenseNet-121, temperature scaling optimizes $T^* \approx 1.0$,
256 producing an uncalibrated T_3 Brier score of 0.1067 compared to raw 0.1079

Table 2: Experimental Matrix Evaluation: Primary Longitudinal Performance across Architectures, Loss Objectives, and Thoracic Pathologies (Strata $T_1 \rightarrow T_3$). Values represent mean $\pm$ standard deviation across three independently trained random seeds ($S \in \{42, 123, 2026\}$). Complete 17-column trajectories across all intermediate deployment strata (T_1, T_2, T_3) and log-loss metrics are detailed in Supplementary Table S3.

Architecture	Objective	Pathology	AUROC ($T_1 \rightarrow T_3$)	Δ AUROC	Brier ($T_1 \rightarrow T_3$)	Δ Brier	ECE T_3
DenseNet-121	Unweighted BCE	Pleural Effusion	0.934 $\rightarrow$ 0.873	-0.060	0.037 $\rightarrow$ 0.028	-0.009	0.0090
DenseNet-121	Unweighted BCE	Cardiomegaly	0.908 $\rightarrow$ 0.902	-0.006	0.053 $\rightarrow$ 0.068	+0.015	0.0210
DenseNet-121	Unweighted BCE	Pneumonia	0.847 $\rightarrow$ 0.865	+0.017	0.024 $\rightarrow$ 0.019	-0.005	0.0095
DenseNet-121	Unweighted BCE	Edema*	0.540 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0029
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	0.906 $\rightarrow$ 0.839	-0.067	0.108 $\rightarrow$ 0.125	+0.017	0.2227
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	0.874 $\rightarrow$ 0.858	-0.016	0.126 $\rightarrow$ 0.136	+0.010	0.1969
DenseNet-121	Pos-Weighted BCE	Pneumonia	0.818 $\rightarrow$ 0.842	+0.024	0.131 $\rightarrow$ 0.130	-0.002	0.2425
DenseNet-121	Pos-Weighted BCE	Edema*	0.612 $\rightarrow$ -	-	0.061 $\rightarrow$ 0.064	+0.002	0.1274
ResNet-50	Unweighted BCE	Pleural Effusion	0.930 $\rightarrow$ 0.861	-0.069	0.037 $\rightarrow$ 0.028	-0.009	0.0132
ResNet-50	Unweighted BCE	Cardiomegaly	0.890 $\rightarrow$ 0.879	-0.011	0.057 $\rightarrow$ 0.071	+0.014	0.0175
ResNet-50	Unweighted BCE	Pneumonia	0.843 $\rightarrow$ 0.880	+0.037	0.024 $\rightarrow$ 0.019	-0.005	0.0106
ResNet-50	Unweighted BCE	Edema*	0.556 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0017
ResNet-50	Pos-Weighted BCE	Pleural Effusion	0.879 $\rightarrow$ 0.819	-0.060	0.126 $\rightarrow$ 0.139	+0.014	0.2751
ResNet-50	Pos-Weighted BCE	Cardiomegaly	0.830 $\rightarrow$ 0.799	-0.031	0.142 $\rightarrow$ 0.152	+0.011	0.2338
ResNet-50	Pos-Weighted BCE	Pneumonia	0.835 $\rightarrow$ 0.823	-0.012	0.167 $\rightarrow$ 0.169	+0.002	0.3253
ResNet-50	Pos-Weighted BCE	Edema*	0.549 $\rightarrow$ -	-	0.069 $\rightarrow$ 0.075	+0.005	0.1670

Table 3: Multi-Calibrator Comparison on Pleural Effusion: Deidentified-Date Ensemble Performance, Calibration Intercept (α), Slope (β), and Paired Bootstrap Contrasts. Complete 12-column factorial evaluation across all intermediate strata and log-loss is provided in Supplementary Table S4.

Architecture	Loss Objective	Calibrator	Brier ($T_1 \rightarrow T_3$)	T_3 α	T_3 β	T_3 ECE	T_3 Δ Brier vs Raw [95% CI]
DenseNet-121	Weighted BCE	Raw (Uncalibrated)	0.092 $\rightarrow$ 0.108	-2.964	1.048	0.2227	-
DenseNet-121	Weighted BCE	Temperature Scaling (T^*)	0.091 $\rightarrow$ 0.107	-2.962	1.029	0.2190	-0.0011 [-0.0014, -0.0009]
DenseNet-121	Weighted BCE	Analytic Offset ($z - \log w$)	0.036 $\rightarrow$ 0.028	-0.445	0.929	0.0126	-0.0795 [-0.0893, -0.0697]
DenseNet-121	Weighted BCE	Platt Scaling ($a + bz$)	0.036 $\rightarrow$ 0.028	-0.229	1.128	0.0176	-0.0798 [-0.0894, -0.0702]
DenseNet-121	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	+0.147	1.044	0.0040	-
ResNet-50	Weighted BCE	Raw (Uncalibrated)	0.110 $\rightarrow$ 0.123	-2.853	1.950	0.2751	-
ResNet-50	Weighted BCE	Temperature Scaling (T^*)	0.088 $\rightarrow$ 0.101	-2.608	1.184	0.1993	-0.0220 [-0.0241, -0.0198]
ResNet-50	Weighted BCE	Analytic Offset ($z - \log w$)	0.040 $\rightarrow$ 0.029	+1.075	1.405	0.0158	-0.0936 [-0.1022, -0.0844]
ResNet-50	Weighted BCE	Platt Scaling ($a + bz$)	0.038 $\rightarrow$ 0.030	-0.191	1.178	0.0223	-0.0930 [-0.1007, -0.0849]
ResNet-50	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	-0.077	1.095	0.0109	-

(Δ Brier = -0.0011 [95% CI: $-0.0014, -0.0009$]) while leaving the calibration intercept severely displaced ($\alpha = -2.9615$ vs. raw -2.9638). For ResNet-50, temperature scaling achieves modest variance shrinkage (Δ Brier = -0.0220 [$-0.0241, -0.0198$]), but leaves the underlying intercept discrepancy uncorrected ($\alpha = -2.6080$ vs. raw -2.8526). Temperature scaling is mathematically incapable of adjusting this intercept displacement.

In contrast, subtracting the theoretical logit offset ($z - \ln w$, with $\ln(21.51) = +3.068$) drops T_3 Brier from 0.1079 to 0.0284 (Δ Brier = -0.0795 [95% CI: $-0.0893, -0.0697$], 73.7% error reduction), reducing ECE from 22.3% to 1.26% and shifting the Cox intercept from $\alpha = -2.9638$ to -0.4451 while

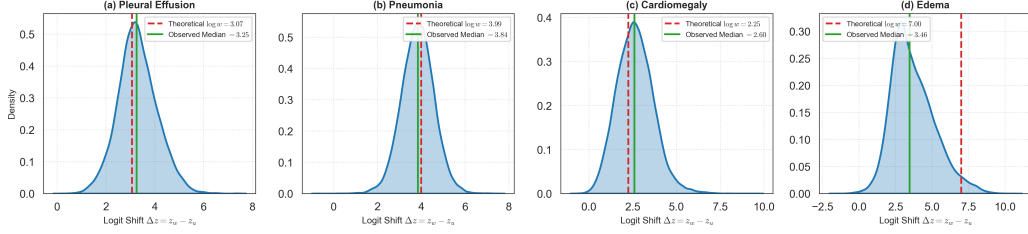


Figure 2: Empirical verification of theoretical logit shift ($z_w - z_u \approx \log w$) across 16,604 held-out deidentified-date image evaluations (8,302 images $\times$ 2 architectures, with Δz computed as the mean difference across the three seed-matched weighted/unweighted model pairs). Kernel density estimates (blue) of observed logit differences between positive-weighted and unweighted models cluster tightly around the theoretical population attractor $\log w$ (dashed red line; observed median in solid green) across Pleural Effusion (a), Pneumonia (b), and Cardiomegaly (c). For Edema (d), extreme training class scarcity ($w > 1000$) induces optimization saturation, bounding empirical shifts near $+3.69$.

perfectly preserving the calibration slope ($\beta = 0.9292$). Fitted Offset ($z + a^*$) and Platt Scaling achieve identical performance (T_3 Brier: 0.0289 and 0.0281; $\Delta\text{Brier} = -0.0790$ and -0.0798). For ResNet-50, Analytic Offset similarly eliminates 76.3% of probability error ($0.1227 \rightarrow 0.0291$, $\Delta\text{Brier} = -0.0936$ [95% CI: $-0.1022, -0.0844$]), confirming that intercept neutralization is architecturally universal.

3.2.1. Empirical Verification of Logit Offset: $\Delta z \approx \ln w$

Evaluating logit differences $\Delta z(x) = \bar{z}_w(x) - \bar{z}_u(x)$ across all 16,604 ensemble evaluations (Fig. 2) confirms that finite neural networks cluster tightly around theoretical $\ln w$: Pleural Effusion ($\ln w = 3.07$) observed mean $\Delta z = 3.29 \pm 0.81$ (median 3.25); Pneumonia ($\ln w = 3.99$) observed 3.83 ± 0.74 (median 3.84); and Cardiomegaly ($\ln w = 2.25$) observed 2.64 ± 1.04 (median 2.60). For Edema ($w = 1091.3, \ln w = 7.00$), acute training scarcity (23 positive cases) causes optimization saturation, bounding empirical shifts at $\Delta z = 3.69 \pm 1.47$ (median 3.46) due to L_2 regularization. Equal-mass reliability diagrams (Fig. 3) demonstrate that the analytic offset realigns curves tightly with the ideal diagonal across all strata.

3.3. Selective Prediction and Workload Triage Simulation

Table 4 and Fig. 4 summarize the selective triage simulation on later held-out stratum T_3 ($N = 2,436$ radiographs, 85 positive Pleural Effusion cases) under development thresholds. At 100% coverage (full automation),

Table 4: Selective Prediction and Reader Referral Triage Simulation on Later Held-Out Stratum T_3 ($N = 2,436$ Radiographs, 85 Positive Pleural Effusion Cases). Evaluated under matched Youden decision thresholds determined on development stratum T_1 across primary clinical operating regimes (100%, 90%, 80%, 70% coverage). Complete 24-row simulation across all coverage levels is detailed in Supplementary Table S5.

Coverage	Referred (Workload)	Strategy	Automated FN (Missed)	Retained Sens.	Full Sens.	Retained AUROC	Retained Brier
100%	0 (0%)	Weighted BCE (Raw)	16.0	81.18%	81.18%	0.8691	0.1079
100%	0 (0%)	Weighted BCE (Analytic Corrected)	16.0	81.18%	81.18%	0.8623	0.0284
100%	0 (0%)	Unweighted BCE (Raw)	12.0	85.88%	85.88%	0.8806	0.0268
90%	243 (10%)	Weighted BCE (Raw)	14.0	82.93%	83.53%	0.8789	0.1062
90%	243 (10%)	Weighted BCE (Analytic Corrected)	15.0	81.01%	82.35%	0.8735	0.0289
90%	243 (10%)	Unweighted BCE (Raw)	12.0	84.62%	85.88%	0.8923	0.0266
80%	487 (20%)	Weighted BCE (Raw)	10.0	87.01%	88.24%	0.8882	0.1012
80%	487 (20%)	Weighted BCE (Analytic Corrected)	15.0	77.61%	82.35%	0.8896	0.0264
80%	487 (20%)	Unweighted BCE (Raw)	12.0	82.61%	85.88%	0.9027	0.0256
70%	730 (30%)	Weighted BCE (Raw)	7.0	89.39%	91.76%	0.8997	0.0956
70%	730 (30%)	Weighted BCE (Analytic Corrected)	13.0	70.45%	84.71%	0.8876	0.0178
70%	730 (30%)	Unweighted BCE (Raw)	11.0	78.00%	87.06%	0.9051	0.0194
70%	730 (30%)	Random Deferral Baseline	11.3	80.99%	86.69%	—	0.1081

the raw weighted model incurs 16.0 missed cases (FN_{auto}) with Brier 0.1079. Calibrated Analytic Offset under matched threshold ($t^* \approx 0.022$) yields identical sensitivity (81.18%) and 16.0 missed cases, but with an almost four-fold lower Brier score of 0.0284. Unweighted raw models achieve 12.0 missed cases (sensitivity 85.88%, Brier 0.0268).

Under selective deferral at 70% coverage (30% referral workload, 730 radiographs referred): raw weighted models halve automated false negatives from 16.0 to 7.0 (retained sensitivity 89.39%, full-cohort triage sensitivity 91.76%), but leave probabilities uncalibrated (Brier 0.0956); conversely, analytic calibrated models achieve superior probability estimation (Brier 0.0178), but retain 13.0 automated false negatives because threshold-distance normalization ($t^* \approx 0.022$) classifies low-probability false negatives ($\hat{p} \ll t^*$) as confident negatives. This reveals a fundamental operational trade-off:

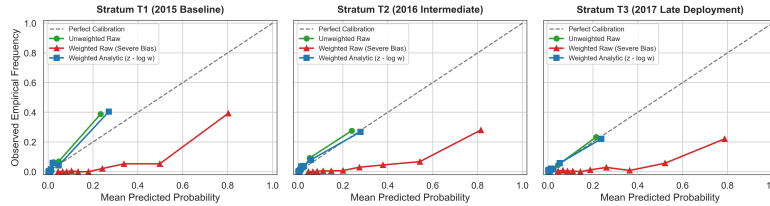


Figure 3: Reliability diagrams for Pleural Effusion (DenseNet-121 3-seed ensemble) across held-out deidentified-date strata (T_1, T_2, T_3). Raw weighted models (red) severely over-predict event probabilities across all bins. Analytic offset calibration (blue) and unweighted models (green) remain tightly calibrated to the ideal diagonal across all deployment windows.

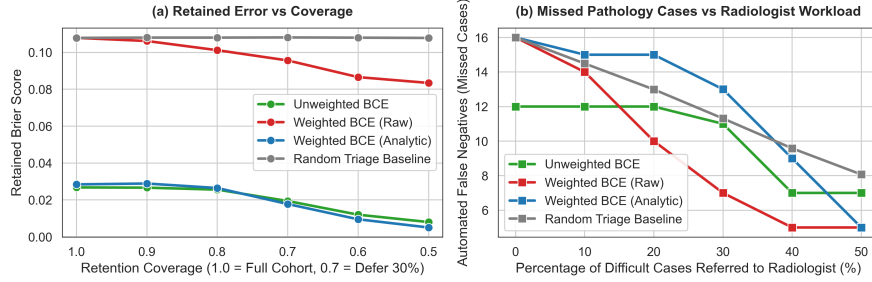


Figure 4: Selective prediction evaluation on later held-out stratum T_3 with matched development thresholds (Table 4). Left: Risk-coverage curves demonstrating that selective deferral steadily reduces Brier error across all models as ambiguous cases are referred. Right: Reader referral workload vs. automated false negatives, illustrating the distinct operational behaviors of raw weighted models (halving missed cases to 7.0 under 30% referral) and analytic corrected models (optimizing probability accuracy with Brier 0.0178).

301 probability calibration is required for valid risk scoring, whereas raw weighted
 302 models with threshold-proximity triage act as a heuristic for recall-oriented
 303 screening. Neither system eliminates missed disease entirely.

304 3.4. Cross-Institutional External Evaluation (MIMIC-CXR-JPG)

305 The external MIMIC-CXR-JPG cohort comprised 3,403 frontal radio-
 306 graphs from 3,041 studies and 289 patients. Rather than asserting turnkey
 307 diagnostic transportability—given moderate discrimination for Pleural Effu-
 308 sion (AUROC 0.688 [95% CI: 0.656, 0.717] DenseNet-121; 0.677 ResNet-50),
 309 weak discrimination for Pneumonia (0.583–0.607) and Edema (0.516–0.581),
 310 and near-chance discrimination for Cardiomegaly (0.519–0.525)—this exper-
 311 iment functions as an external stress-test of zero-retraining recalibration
 312 under domain shift. Negative Brier skill scores (BSS < 0; Table 5) confirmed
 313 substantial cross-institutional baseline calibration shift relative to a local null
 314 model ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$).

315 Paired analytic recalibration ($\bar{p}_{\text{corr}} = \frac{1}{3} \sum \sigma(z_s - \ln w)$) demonstrated sta-
 316 tistically significant Brier reductions for Cardiomegaly ($\Delta\text{Brier} = -0.0579$
 317 [95% CI: $-0.0790, -0.0349$] DenseNet; $-0.0611 [-0.0818, -0.0415]$ ResNet)
 318 and Pneumonia ($\Delta\text{Brier} = -0.0095 [-0.0170, -0.0016]$ DenseNet; -0.0155
 319 $[-0.0254, -0.0059]$ ResNet), with point-estimate reductions on Pleural Ef-
 320 fusion (-0.0084 and -0.0109). For Edema, subtracting theoretical $\ln w$
 321 worsened Brier score ($+0.0217$), consistent with over-correction when empiri-
 322 cal finite-network shifts saturate below theoretical $\ln w$. Diagnostic study-level

Table 5: Cross-Institutional External Validation on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Images from 3,041 Studies Across $N = 289$ Unique Patients). Primary experimental evaluation of paired zero-parameter analytic logit recalibration ($\bar{p}_{\text{raw}} = \frac{1}{3} \sum \sigma(z_s) \rightarrow \bar{p}_{\text{corr}} = \frac{1}{3} \sum \sigma(z_s - \log w)$) on 3-seed ensembles trained with positive-weighted BCE. Evaluated with empirical prevalence (π), local null reference baseline ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), patient-clustered 1,000-replicate bootstrap 95% confidence intervals, Brier skill score ($\text{BSS} = 1 - \text{Brier}/\text{Brier}_{\text{null}}$), and paired contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$). The complete 16-row factorial comparison with unweighted BCE baselines is detailed in Supplementary Table S1, and study-level sensitivity analysis is detailed in Supplementary Table S2.

Architecture	Pathology	Prevalence (π)	Brier _{null}	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

sensitivity analysis ($N = 3,041$ studies across 289 patients, maximum probability pooling; Supplementary Table S2) demonstrated identical concordance, confirming that recalibration patterns hold across radiographs and studies.

4. Discussion

4.1. Decoupling Loss Distortion from Deidentified-Date Cohort Shift

Our findings clarify the mechanism behind reported “temporal calibration decay” in medical AI [8, 9]. While moderate discriminatory attenuation does occur across subsequent cohort strata (AUROC drops by ≈ 0.06 identically across unweighted and weighted models), the perception of progressive calibration collapse is largely an artifact of positive loss weighting interacting with shifting disease prevalence. Because standard post-hoc evaluation relies almost exclusively on temperature scaling—which modifies logit dispersion without shifting the intercept—evaluators have routinely conflated static objective-loss bias with intrinsic temporal feature drift. When the mathematical logit offset is neutralized ($z - \ln w$), the apparent calibration collapse vanishes, and Brier scores track the unweighted baseline across all held-out strata.

4.2. Relationship to Prior Theoretical and Clinical Literature

Our theoretical framing connects with the work of Caplin, Martin, and Marx [14], who demonstrated that class weighting corrupts calibration and derived an inverse likelihood recovery framework on static single-split tasks. We

344 advance this literature into longitudinal clinical deployment: (i) resolving the
 345 temporal calibration decay paradox across sequential deidentified-date cohort
 346 strata; (ii) proving the structural failure of temperature scaling ($\alpha \approx -2.96$);
 347 (iii) demonstrating finite-network optimization bounds ($w > 1000$ in Edema
 348 saturates at $\Delta z \approx 3.69$ vs. theoretical 7.00); and (iv) stress-testing cross-
 349 institutional transfer from BRAX to MIMIC-CXR. Furthermore, extending
 350 multi-dataset selective CXR investigations [20], we operationalize the clinical
 351 triage trade-off: analytic offset ($z - \ln w$) is essential for epidemiological risk
 352 stratification and prognosis (restoring Brier parity, consistent with plug-in
 353 principles [17]), while raw weighted models support recall-oriented screening
 354 triage under referral budgets.

355 4.3. Architectural Universality and Operational Utility

356 The observed phenomena are consistent across architectures: DenseNet-
 357 121 and ResNet-50 exhibit identical discrimination attenuation ($\Delta \text{AUROC} =$
 358 -0.0604 vs. -0.0687), severe raw calibration intercepts ($\alpha \approx -2.96$ and
 359 -2.85), and comparable Brier reductions under Analytic Offset (73.7% and
 360 76.3%, $\Delta \text{Brier} = -0.0795$ and -0.0936). Because Analytic Offset ($z - \ln w$)
 361 operates in constant time ($O(1)$) with zero validation fitting and zero refitting,
 362 it provides an immediate algorithmic solution for resource-constrained edge
 363 devices and PACS pipelines.

364 4.4. Scope, Boundaries, and Methodological Limitations

365 Several methodological limitations define the boundary of these findings:

- 366 1. **Date Anonymization and Temporal Semantics:** In BRAX, dates
 367 are fictitious hashes preserving intra-patient acquisition intervals; cross-
 368 patient calendar synchronization is unverified. Partitions (T_1, T_2, T_3)
 369 reflect sequential deidentified-date strata rather than synchronized cal-
 370 endar years.
- 371 2. **Cross-Institutional Domain Gap and Modest Transportability:**
 372 External validation on MIMIC-CXR-JPG yielded modest discrimina-
 373 tion (Effusion AUROC ≈ 0.69) and near-chance discrimination for
 374 Cardiomegaly (0.52). While analytic offset reduced Brier error for
 375 Cardiomegaly and Pneumonia, it worsened Edema (+0.0217), demon-
 376 strating that zero-retraining offset correction cannot compensate for
 377 severe cross-institutional covariate shift or negative transfer.

- 378 **3. Label Extraction Noise and Retrospective Simulation:** La-
 379 bels were derived from Portuguese reports via NLP rather than multi-
 380 radiologist panel review, and evaluations were retrospective. Real-world
 381 validation requires prospective silent shadow deployment in operational
 382 hospital PACS.
- 383 **4. Architectural Scope and Scarcity Bounds:** We evaluated standard
 384 CNN backbones (DenseNet-121, ResNet-50); Vision Transformers and
 385 vision-language models warrant future investigation. Under extreme
 386 class scarcity (Edema, $w = 1091.3$), empirical logit shifts saturate at
 387 $+3.69$ due to L_2 regularization, defining the empirical boundary of the
 388 asymptotic derivation.

389 5. Conclusion

390 In this controlled study across 40,523 BRAX radiographs, severe probabil-
 391 ity miscalibration across held-out strata was predominantly driven by positive
 392 loss weighting rather than progressive temporal drift. Because standard
 393 temperature scaling cannot shift the logit intercept, it leaves this distortion
 394 intact. A constant-time closed-form analytic offset ($z - \ln w$) eliminates
 395 73.7% of stratum T_3 Brier error ($0.1079 \rightarrow 0.0284$, $\Delta\text{Brier} = -0.0795$ [95%
 396 CI: $-0.0893, -0.0697$]) without model refitting or validation data, restoring
 397 calibration parity with unweighted models. Furthermore, selective triage
 398 simulations reveal an operational trade-off: raw weighted models provide
 399 effective recall-oriented triage by halving missed cases under a 30% refer-
 400 ral budget, whereas analytic calibration restores reliable probabilities for
 401 decision-theoretic risk stratification. Neither approach eliminates missed
 402 disease entirely, underscoring the necessity of transparent, workload-aware
 403 deployment policies in clinical deep learning.

404 CRediT Authorship Contribution Statement

405 **Vishesh Panghal:** Conceptualization, Methodology, Software, Validation,
 406 Formal Analysis, Investigation, Data Curation, Writing – Original Draft,
 407 Writing – Review & Editing, Visualization, Project Administration.

408 Declaration of Competing Interest

409 The author declares that he has no known competing financial interests
 410 or personal relationships that could have appeared to influence the work

411 reported in this paper.

412 Data and Code Availability

413 The BRAX dataset (v1.1.0) [2] is available on PhysioNet at <https://physionet.org/content/brax/1.1.0/> under the PhysioNet Credentialed
414 Data Use Agreement. The MIMIC-CXR-JPG database (v2.1.0) [21, 22] is
415 available on PhysioNet at <https://physionet.org/content/mimic-cxr-jpg/2.1.0/> under the PhysioNet Credentialed Data Use Agreement. Access re-
416 quires credentialed PhysioNet researcher status, completion of human subjects
417 research training, and signed data use agreements. Complete end-to-end repli-
418 cation scripts, model evaluation routines, cross-fitting recalibration pipelines,
419 selective prediction triage simulations, and tabular evaluation artifacts are
420 openly available on GitHub at [https://github.com/Vishesh-panghal/](https://github.com/Vishesh-panghal/temporal-brax-calibration)
421 [temporal-brax-calibration](https://github.com/Vishesh-panghal/temporal-brax-calibration), and the repository will be permanently as-
422 signed a persistent DOI via Zenodo upon acceptance of the manuscript. No
423 credentialed patient-level images, metadata, labels, identifiers, or prediction
424 rows are redistributed; the repository contains aggregate evaluation artifacts
425 and a synthetic manifest schema example.
426
427

428 Ethical Approval and Consent to Participate

429 This study represents a secondary analysis of credentialed-access, de-
430 identified diagnostic imaging databases accessible under formal Data Use
431 Agreements. The primary BRAX dataset collection was approved by the
432 Institutional Review Board (IRB) of Hospital Israelita Albert Einstein (São
433 Paulo, Brazil; CAAE: 36720520.1.0000.0071) with a formal waiver of informed
434 consent. Creation and distribution of the MIMIC-CXR-JPG database was
435 approved by the Institutional Review Boards of Beth Israel Deaconess Medical
436 Center (Boston, MA) and the Massachusetts Institute of Technology (Protocol
437 #0403000206) with a waiver of informed consent. The author completed
438 mandatory human subjects research training (CITI Program certification),
439 executed approved PhysioNet Data Use Agreements for both databases, and
440 strictly complied with data governance protocols prohibiting re-identification
441 or redistribution of protected health information.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author utilized generative artificial intelligence platforms (Google Gemini, Anthropic Claude, and OpenAI Codex) exclusively for LaTeX formatting and syntax verification, analysis code refactoring and modularization, grammatical refinement, and bibliographic cross-referencing. Following the use of these tools, the author thoroughly reviewed, verified, and edited all text, numerical outputs, and code, and takes full intellectual and scientific responsibility for the integrity, validity, and conclusions of the published manuscript.

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